Rest Is Not Armor

Does an ace-every-6th-day rotation protect the 2027 Dodgers staff?

MLB 2010-2026 | October 29, 2026

The question. This is a hypothetical: how I would run the 2027 Dodgers' rotation. My plan: Yoshinobu Yamamoto starts every sixth day no matter what (the seventh if the sixth is an off day), and the rest of the rotation fills every other game. Adrian Morejon, a career reliever with the Padres, joins the Dodgers as a hypothetical free-agent signing. I tested three versions: ace + 6 with Morejon; ace + 5 with Morejon starting and Wrobleski in the bullpen; and ace + 5 the other way around. Does any of them keep arms healthier, and what does each cost?

The short answer. Not by much. Across 72,391 starts since 2010, I found that rest beyond the normal four days does little for arm health. A fifth day of rest barely moves arm-injury risk, and a sixth helps only modestly and not reliably. Rest doesn't change how well a starter pitches either. The every-6th-day rule works for Yamamoto: about 30 starts, none on four days of rest. Of the three versions, ace + 5 with free agent Morejon starting and Wrobleski in the bullpen projects the most runs, and ace + 6 with Morejon is the safest. Morejon is the bet: past relievers who moved into a rotation kept their run value, but his arm history makes him the staff's biggest injury risk.

Key findings

1. **A fifth day of rest barely matters.** When the schedule alone decides whether a pitcher gets 4 or 5 days (an off day fell inside his turn), arm-IL odds are 11% lower with five days (95% CI 0.71-1.11, 2017-26). Over all years it is 2% lower.
2. **A sixth day may help, but the evidence is thin.** Six days points to 21-31% lower odds than four, depending on the design. But only 19% of starts come on 6+ days, so none of the intervals rule out zero.
3. **Rest doesn't change performance.** Comparing the same pitcher within a season, velocity, K-BB%, wOBA allowed and run value hardly move between 4, 5 and 6+ days. Extra rest mainly makes outings slightly shorter: 8-9 days costs about 1.6 pitches.
4. **Recent injuries and workload jumps matter more than rest.** Passing last season's pitch total raises arm-IL odds 27% (p 0.1). That matters for this staff: Morejon hasn't started since 2023, and Ohtani, Snell and Glasnow all threw well under 1,400 MLB pitches in 2026.
5. **Ace + 5 with Morejon starting projects the most runs.** With Wrobleski in the bullpen, it projects +58 starter runs net of injuries. Ace + 6 with Morejon projects +52, and ace + 5 with Wrobleski starting +45. Its injury exposure (2.39 expected arm-IL stints) is about what a strict 6-man gives.
6. **The free-agent bet on Morejon is reasonable but risky.** 22 relievers who moved into a rotation since 2021 lost 1.4 mph and about 2.7 points of K-BB%, but kept their run value (+0.35 to +0.36 per 100 pitches). His arm history gives him the highest per-start arm risk on the staff: 0.31 expected arm-IL stints in 22 starts under ace + 6, 0.40 in 26 under ace + 5, and he passes his 2026 pitch total around August 6.
7. **My plan: ace + 6 with Morejon building up behind Sheehan.** Morejon opens at 45, 55, 62 and 70 pitches, and Sheehan follows the three starts under 65 for 35 pitches each (April 1-17), always on at least 3 days of rest. It costs about 0.4 runs against plain ace + 6. It takes about 4 April relief outings from Sheehan and keeps his season near 1,016 pitches, about half his 2026 load.
8. **Simulated over 2,000 seasons with the 2026 offense, my plan wins about 99 games.** The 10th-90th percentile range is 91-107. Depth starters (mostly Stone) cover about 22 starts. The seven-man bullpen takes about 6.5 IL stints a season, with heavy back-to-back use for the setup men.
9. **Head to head in the simulation, the gaps are small.** Wins run from 98.4 to 100.5 across five plans. The 6-man, ace + 5 and a strict 5-man without Morejon edge out my ace + 6 by 0.5-1.3 wins, mostly because seven starters leave only seven relievers. A AAA shuttle with optionable arms halves my plan's short-handed outings. Ace + 6 with Sheehan instead of Morejon is the weakest.
10. **Starting Yamamoto on his 5th day before an off day adds about 0.4 wins.** He gets 32 starts instead of 30, 7 of them on four days of rest, taken from Morejon and Wrobleski. His expected arm-IL stints rise from 0.37 to 0.42, mostly because he passes his 2026 pitch total sooner, not because of the short rest.

Data: 72,391 regular-season starts by 903 rotation pitchers, 2010-2026 (Retrosheet through 2025, Statcast for 2026), with 789 arm injured-list placements that followed a start. IL placements come from MLB's transactions feed. Methods are in the appendix.

PART 1

The league already gives more rest, and arms still break

Before testing rest start by start, here is the big picture. Clubs have been giving starters more rest for 15 years. In 2010, 51% of starts came on four days of rest; in 2026, 30%. Pitch counts fell too. Arm injuries went the other way.

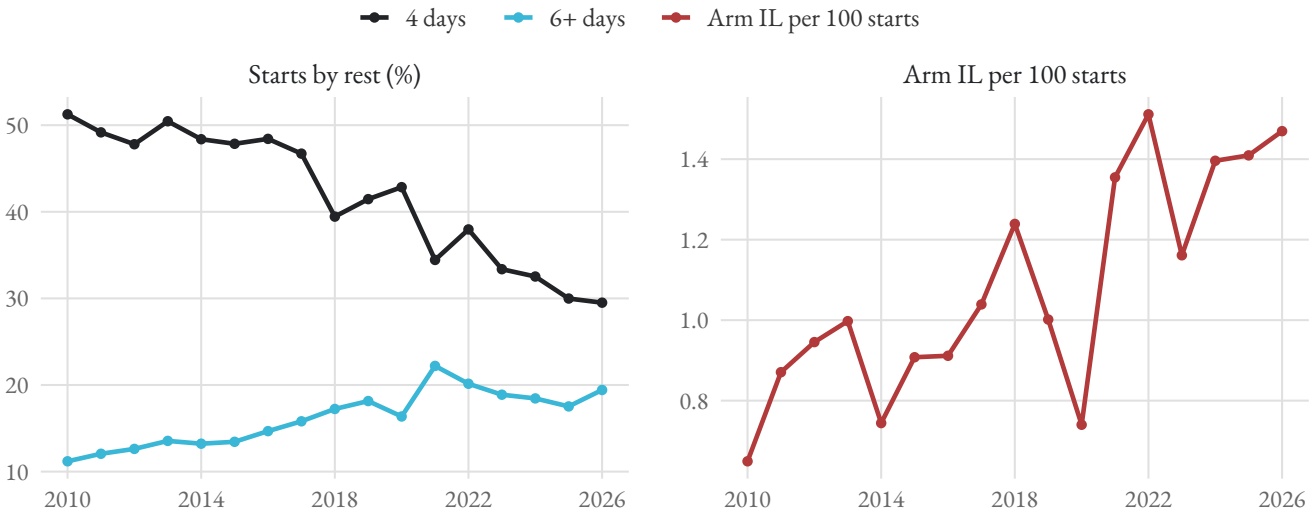


Figure 1: Rotation starters by season. Left: share of starts on 4 days of rest and on 6+ days. Right: arm IL placements per 100 starts (placements effective before the pitcher's next appearance). 2020 is a 60-game season.

That doesn't prove rest is useless. Velocity and spin also rose, and IL rules changed (the 10-day IL arrived in 2017). But it sets expectations: the league as a whole moved toward more rest and saw no drop in arm injuries.

> **Takeaway.** The league already moved a long way toward more rest, and arm-IL rates rose. If rest were strong protection, that would be hard to explain.

PART 2

Does an extra day of rest prevent arm injuries?

The hard part is cause and effect. Clubs give extra rest to pitchers who feel something, so comparing rested starts with short-rest starts mixes rest with soreness. The cleanest comparison uses the schedule. About 69% of starts come exactly five team games after the pitcher's last one. For those, whether he got 4, 5 or 6 days depended only on whether an off day fell inside his turn, not on how his arm felt.

The outcome is an arm IL placement (elbow, shoulder, forearm, biceps/triceps, lat) effective after the start and before the pitcher's next appearance (at most 21 days). Every model controls for age, injury history, the previous start's pitch count, month and season.

- **Four vs. five days: no real difference.** With schedule-set rest, raw rates are 1.13 arm IL placements per 100 starts on four days and 1.15 on five. Every design puts the odds ratio between 0.89 and 0.98.
- **Six days leans safer.** The odds ratios run 0.69-0.79. Only one design (all starts) comes close to significance (p 0.073), and that design is the one most exposed to clubs' choices.
- **A 6-man turn adds nothing by itself.** With rest held at five days, starts on a 6-man turn are no safer than starts on

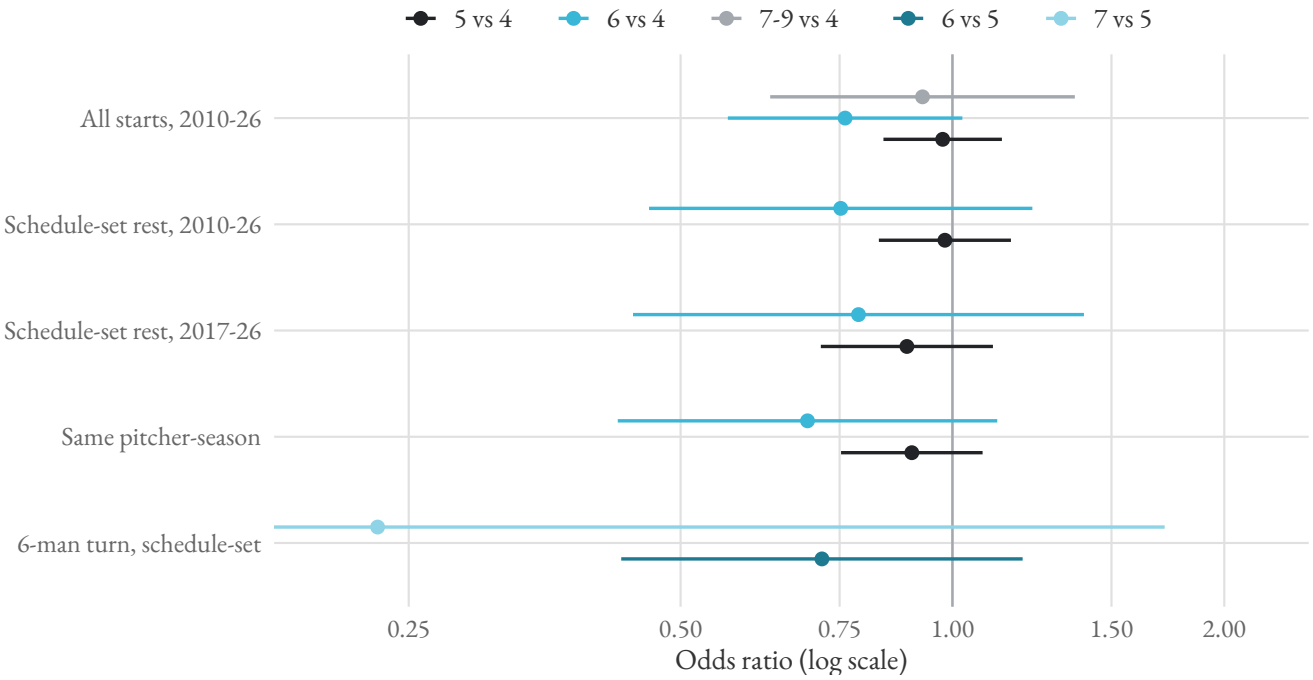


Figure 2: Odds of an arm IL placement after the start by days of rest, relative to 4 days (5 days for the 6-man turn). Below 1 means safer. Bars are 95% intervals clustered by pitcher. Schedule-set: on-turn starts where off days alone set the rest. Same pitcher-season: conditional logit within each pitcher's season.

a 5-man turn with an off day (OR 1.15, p 0.39).

- **Repeated 4-day turns aren't a hidden risk.** Across all starts, a higher share of four-day rests over the last five starts looks risky (OR 1.60 from none to all). When the schedule sets that share, the effect shrinks to 1.13 (p 0.61).

> **Takeaway.** The fifth day of rest, which is what a 6-man rotation buys, does almost nothing for arm health. Six or more days may cut risk by roughly a fifth, but the data can't confirm it.

PART 3

Does rest change how well he pitches?

If rest doesn't prevent injuries, it might still help performance. These comparisons look within each pitcher's season, so they ask whether the same pitcher throws better on five days than on four. Velocity and run value come from Statcast (2020-26); runs and K-BB% cover 2010-26.

- **Four days isn't worse.** On four days, fastballs are +0.02 mph and K-BB% is +0.26 points, both tiny and if anything *better* than on five days. Run value and wOBA allowed don't move.
- **Long rest doesn't sharpen anyone.** Six to nine days shows no velocity or run-value gain. The only clear effect is shorter outings: −1.6 pitches and −0.37 outs at 8-9 days. That is probably managers easing pitchers back after a layoff, not rust.
- **The season doesn't wear velocity down either.** Within a season, fastball velocity is lowest in the first 500 pitches (−0.08 mph against the season average) and flat after that, including past 2,500 pitches.

> **Takeaway.** Rest is not a performance tool. A pitcher on four days pitches like himself on five, and extra days don't make him sharper.

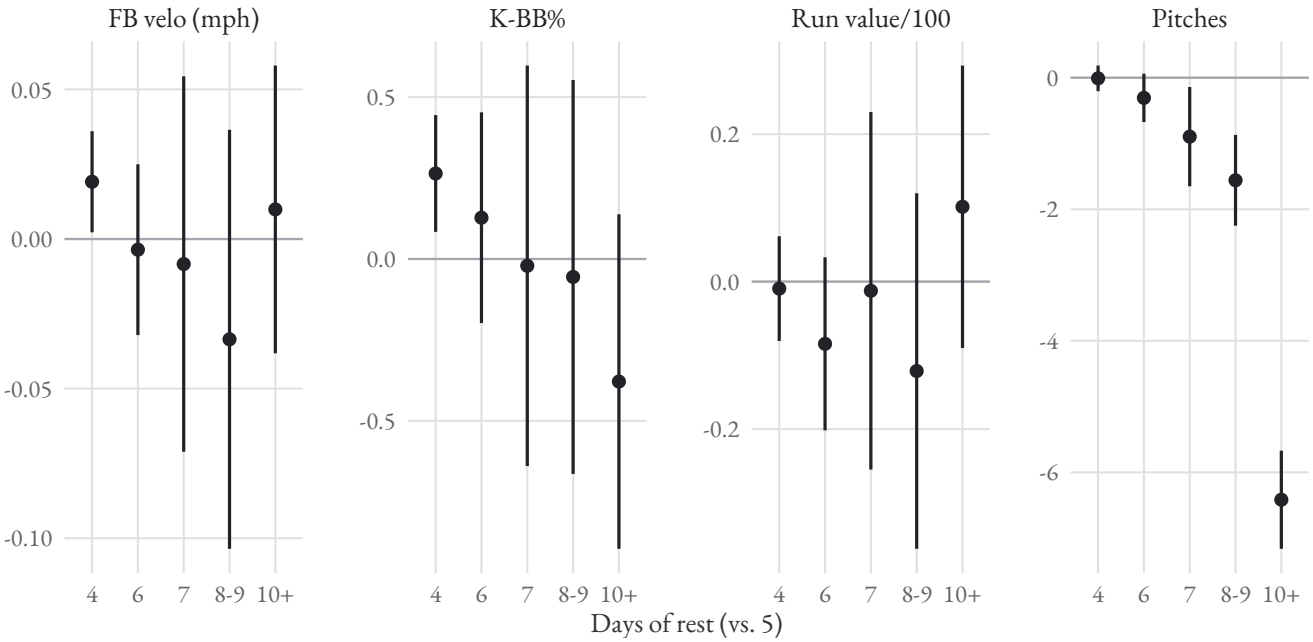


Figure 3: Change against the same pitcher's starts on 5 days of rest, same season, controlling for month. All starts by rotation pitchers; starts followed by an arm IL are dropped. Bars are 95% intervals clustered by pitcher. Run value is from the pitcher's side (positive = better).

PART 4

What does predict arm injuries?

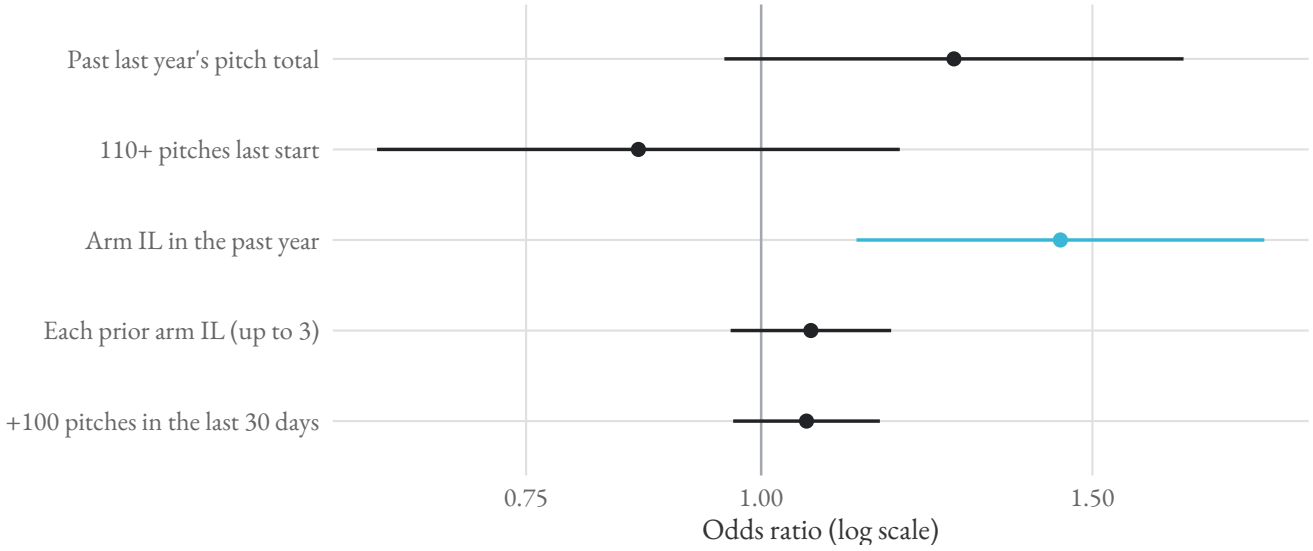


Figure 4: Arm IL odds ratios for other workload and history factors. 95% intervals clustered by pitcher.

- **Workload jumps.** After a pitcher passes his previous season's MLB pitch total, his arm-IL odds per start are 27% higher. In the 2018-26 model used for the projections the jump is larger (about 48%, p 0.002). Starts past 120% of last year's total run 1.12 per 100, against 0.86 at 80-100%.
- **Recent arm injury.** An arm IL stint in the past year raises the odds 44%. Each earlier arm stint adds a little more.
- **Not high pitch counts.** A 110+ pitch previous start doesn't predict injury (OR 0.86). Neither does the 30-day pitch

load. Low pitch counts look "risky" only because pitchers who are already hurting get pulled early.

- **Not last season's volume.** Among pitchers with 20+ starts, a heavier season doesn't raise next season's arm-IL odds (OR 1.05 per 1,000 pitches). Pitchers who make every start tend to be the durable ones.

> **Takeaway.** The risk factors that matter are a recent arm injury and a big jump over last year's workload, not short rest or high pitch counts. Both apply to this staff.

PART 5

The 2027 Dodgers: four ways to build around Yamamoto

Every version keeps my rule: Yamamoto starts every sixth day (the seventh after an off day) and never on four days of rest. The one exception is a test of my build-up plan where he goes on the fifth day instead when his sixth is an off day. The versions differ in who fills the other games. **Adrian Morejon is a free-agent signing.** He has made only 9 MLB starts, all short openings (64 pitches at most) and none since 2023, and he worked purely in relief for the Padres from 2024 to 2026. I show strict 6-man and 5-man rotations for reference.

Table 1: The plans. FA = free-agent signing.

Plan	Around Yamamoto	Bullpen
Ace + 6 (Morejon ramp + Sheehan)	Same as ace + 6; Morejon (FA) builds up 45/55/62/70, Sheehan follows starts under 65	Sheehan (7th man)
Ace + 6 (ramp, Yamamoto early)	Same as the build-up plan; Yamamoto goes on his 5th day when his 6th is an off day	Sheehan (7th man)
Ace + 6 (Morejon)	Snell, Ohtani, Glasnow, Sasaki, Morejon (FA), Wrobleski	
Ace + 5 (Morejon starts)	Snell, Ohtani, Glasnow, Sasaki, Morejon (FA)	Wrobleski
Ace + 5 (Wrobleski starts)	Snell, Ohtani, Glasnow, Sasaki, Wrobleski	Morejon (FA)
Strict 6-man	Strict turn: Yamamoto, Snell, Ohtani, Glasnow, Sasaki, Morejon (FA)	Wrobleski
Strict 5-man	Strict turn: Yamamoto, Snell, Ohtani, Glasnow, Sasaki	Wrobleski (Morejon not signed)

I ran each plan through the actual 2027 schedule (162 games, 24 off days). For every start the model gives an arm-IL risk from the pitcher's age, arm-injury history, recent stints, how far he is past his 2026 pitch total, the month, and his days of rest. The rest effect uses the schedule-set 2017-26 estimates (0.89 for five days and 0.79 for six or more). Starter run value is each pitcher's 2024-26 run value per 100 pitches, shrunk toward average, times his expected pitches. Morejon keeps his shrunk relief value (+0.67); other cases are tested below. An arm-IL stint is expected to cost 65 days, and a replacement-level starter covers the missed starts.

Table 2: 2027 plans. Top-3 share: starts by Yamamoto, Ohtani and Snell. Expected arm-IL stints for the staff, with the range from a fitted rest effect to no rest effect. Starter runs: run value above average across all starts. Net of injuries: after missed starts go to a replacement-level starter.

Plan	Yamamoto GS	Top-3 share	Exp. arm IL	Range	Starts lost	Starter runs	Net of injuries
Ace + 6 (Morejon ramp + Sheehan)	30	46%	2.21	1.91-2.70	18	+71	+52
Ace + 6 (ramp, Yamamoto early)	32	47%	2.23	1.94-2.70	19	+72	+52
Ace + 6 (Morejon)	30	46%	2.22	1.91-2.71	18	+71	+52
Ace + 5 (Morejon starts)	30	52%	2.39	2.22-2.86	23	+82	+58
Ace + 5 (Wrobleski starts)	30	52%	2.33	2.15-2.78	22	+67	+45
Strict 6-man	27	50%	2.37	2.20-2.88	22	+80	+57
Strict 5-man	33	60%	2.75	2.59-2.96	31	+80	+48

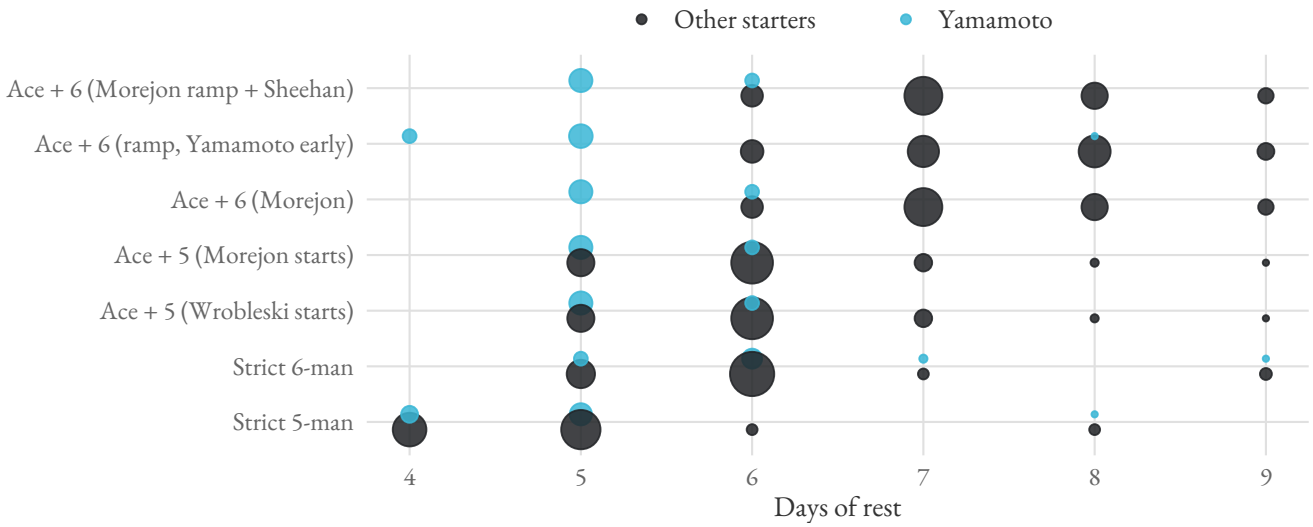


Figure 5: Days of rest for every 2027 start under each plan (first starts and the All-Star break left out). Dot size shows how many starts.

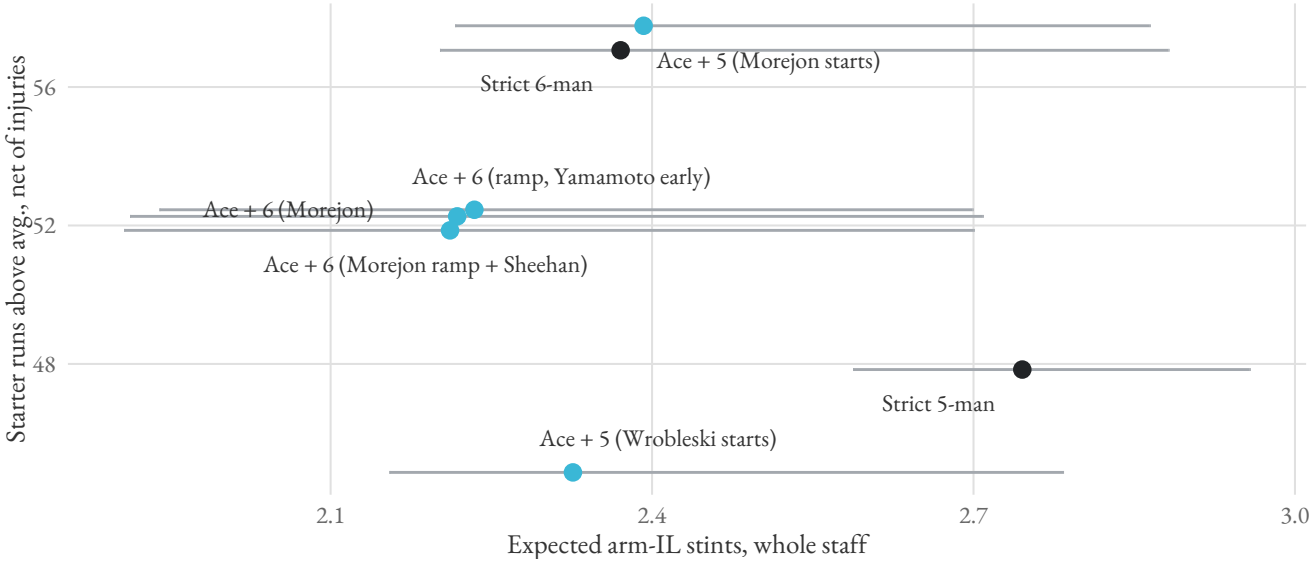


Figure 6: Trade-off by plan: expected staff arm-IL stints (bars show the range across rest-effect assumptions) against starter runs above average after injuries. Cyan: my plans. Charcoal: strict rotations for reference.

- **Ace + 6 with Morejon: safe, but the sixth and seventh starters cost innings.** It expects 2.22 arm-IL stints, 0.15 fewer than a 6-man. That comes mostly from Snell, Ohtani and Glasnow making 22 starts instead of 27, not from extra rest. Net of injuries it projects +52, about 5 runs behind a 6-man, because Morejon and Wrobleski make 44 starts.
- **Yamamoto early: two more ace starts for a small risk.** Going on his 5th day before an off day gives Yamamoto 32 starts instead of 30, 7 of them on four days of rest, and Morejon and Wrobleski lose one start each. Staff arm-IL stints go from 2.21 to 2.23 (Yamamoto's own from 0.37 to 0.42, mostly from passing his 2026 pitch total sooner). Starter runs net of injuries go from +51.9 to +52.5. Part 7 plays it out over full seasons.
- **Ace + 5 with Morejon starting: the most runs.** With Wrobleski in the bullpen, the five others average 5.9 days of rest with none on four, and the plan projects +58, the best of the three and level with a strict 6-man. The price is injury exposure: 2.39 stints, the most of my three plans.
- **Ace + 5 with Wrobleski starting: the weakest of the three.** Keeping Wrobleski in the rotation and Morejon in the bullpen projects +45. Its injury exposure (2.33) is middling. Wrobleski's starter run value is below average (−0.09

per 100 pitches), so 26 starts from him cost runs. The model also gives no credit for Morejon's bullpen value, which partly offsets this in real life.
- **These gaps are small next to the uncertainty.** Every plan expects two to three arm-IL stints, and run differences of a few runs are within the error in the pitcher-quality inputs.

Morejon (free agent) as a starter

Morejon's 9 career starts (2019-23) were all short openings of 64 pitches or fewer. From 2024 to 2026 he was worth +1.25 runs per 100 pitches in relief for the Padres, and he averaged 99.4 mph in 2026. His longest outing was 47 pitches. To see what conversion usually costs, I found every pitcher since 2021 with 35+ relief outings and three or fewer starts in one season, then 10+ starts the next.

- **22 conversions (median 20 starts).** Fastballs lost 1.4 mph and K-BB% fell from 16.1 to 13.4. Run value per 100 pitches held (+0.35 in relief, +0.36 as starters).
- **Relief quality didn't carry over pitcher by pitcher.** A converter's relief run value barely predicted his starter run value (slope 0.02). Only one converter came in as an elite reliever, so whether Morejon keeps an elite edge is open.
- **Injury risk was about normal.** 32% of converters had an arm IL stint in the conversion year, against 29% for all starters with 10+ starts.
- **His own risk is the real issue.** Morejon had five arm IL stints from 2019 to 2023. As a 26-start starter the model gives him 0.40 expected arm-IL stints, a 33% chance of at least one, the highest per-start risk on the staff. At 75 pitches a start he passes his 2026 total (1,344) at start 19, around August 6, when his risk per start rises again.

Table 3: Starter runs above average, net of injuries, under four assumptions for free agent Morejon as a starter (runs per 100 pitches in parentheses). Plans without Morejon in the rotation don't change.

Plan	Penalty (+0.07)	Avg. converter (+0.35)	Shrunk relief (+0.67)	Raw relief (+1.25)
Ace + 5 (Morejon starts)	+48	+52	+58	+68
Ace + 6 (Morejon)	+43	+48	+52	+61
Ace + 5 (Wrobleski starts)	+45	+45	+45	+45
Strict 6-man	+47	+52	+57	+67

- **Ace + 5 with Morejon starting stays on top** of my three plans under every assumption, even the harshest penalty. Ace + 6 with Morejon beats ace + 5 with Wrobleski starting unless Morejon takes the full penalty.

My plan: Morejon's build-up with Sheehan behind him

I'm going forward with ace + 6 and free agent Morejon. Since he hasn't thrown more than 47 pitches in a game since 2024, he builds up over his first four starts: 45, 55, 62 and 70 pitches, then 75 a start. That sits a little below how the 22 past converters ramped (median 60, 73, 70, 80). Any start I plan under 65 pitches has **Emmet Sheehan**, the seventh man in the bullpen, follow for 35 pitches.

Table 4: Morejon's first six starts. Rest of pen: pitches left for the other relievers in an average game (148 team pitches in 2026).

Start	Date	Morejon	Sheehan	Rest of pen	Rest of pen, plain ace + 6
1	Apr 01	45	35	68	73
2	Apr 09	55	35	58	73
3	Apr 17	62	35	51	73
4	Apr 25	70		78	73
5	May 04	75		73	73
6	May 12	75		73	73

Sheehan's rest and bullpen usage. His rest rules come from how 2023-26 relievers were actually used. After 30+ pitches the median reliever sat 3 days before his next outing, after 20-29 pitches 2 days, and after fewer than 20, 1 day. I

give Sheehan the same rules, is never used three days running, and has nothing scheduled in the 3 days before a follow so he's fully rested for it. On other days he works like a typical long reliever: about 40 outings of 26 pitches over a season. I simulated the season 4,000 times with and without the follow assignments.

Table 5: Sheehan's projected 2027 season (mean, with the 10th-90th percentile range across simulations).

Role	Outings	Relief outings	Follow outings	Pitches
Follows Morejon's build-up + long relief	39 (35-43)	35.8	3	1,016 (901-1,130)
Long relief only	40 (36-44)	40.1	0	1,025 (906-1,145)

- **The follow dates are April 1 through April 17.** Morejon's first three starts fall eight days apart, so Sheehan follows on at least 3 days of rest every time (usually about a week).
- **The cost is about 4 relief outings, all in April.** Holding him back before each follow and resting him after takes him out of the bullpen around those dates. By May he is a normal long man again.
- **His workload drops a lot.** About 1,016 pitches against 1,960 as a starter in 2026, so there is no workload jump. He is two years past Tommy John surgery, and 35-pitch follows on a week of rest are a light job.
- **Morejon's build-up costs almost nothing in runs.** The four short starts take about 0.4 runs off his value. Sheehan's 35 pitches are about league average (+0.03 per 100 pitches as a starter, shrunk). Across those starts the rest of the bullpen throws 37 fewer pitches than in plain ace + 6.
- **The ramp delays Morejon's workload jump but barely moves his projected risk.** He passes his 2026 pitch total at start 20 instead of start 19, and his expected arm-IL stints stay at 0.31 either way (22 starts). The model can't credit a gradual build-up directly, because the history has no clean measure of how a ramp itself changes injury risk. Its real value is getting a career reliever stretched out without asking him for 75 pitches on day one.

Why Morejon starts and Sheehan follows

I put Morejon first on purpose. The projection values both pitchers by pitches times run value, so it gives the same answer either way, but the order changes which hitters each one faces. Lineups are built against the starter's hand. In 2026, first-inning lineups were 71% right-handed hitters against a left-handed starter and 39% against a right-handed starter.

Table 6: wOBA allowed by batter hand, 2024-26 (lower is better for the pitcher).

Pitcher	vs LHH	vs RHH
Morejon (LHP)	.245 (364 PA)	.296 (527 PA)
Sheehan (RHP)	.324 (364 PA)	.274 (397 PA)

Table 7: Expected wOBA allowed against the lineup each order invites, weighted by pitches in the three build-up starts.

Order	First pitcher	Follower	Combined
Morejon starts, Sheehan follows	.281	.289	.284
Sheehan starts, Morejon follows	.305	.265	.281

- **The follower gets the platoon edge either way.** With Morejon starting, the opponent stacks right-handed hitters. Morejon faces his weaker side, and Sheehan follows into a lineup full of righties, his strong side. Flip the order and the same thing happens in reverse.
- **On run prevention it's a wash.** The combined difference is about .004 wOBA, under a tenth of a run per game across three games. Pinch-hitters after the switch would shrink it further.
- **The order matters for the build-up.** If Sheehan started, Morejon would just be doing his old long-relief job. Starting gives Morejon a starter's pregame routine, the first inning, and turns on a fixed schedule, which is what converting

a reliever needs. Sheehan, who made 22 starts in 2026, is the natural fit to come in for 35 pitches behind him.

> **Takeaway.** The every-6th-day rule works for Yamamoto in every version. My plan, ace + 6 with free agent Morejon building up behind Sheehan, costs almost nothing against plain ace + 6 (−0.4 runs). It eases Morejon in, spares the rest of the bullpen in April, and gives Sheehan a light, well-rested role. On paper ace + 5 with Morejon starting projects a few more runs, but the gap is within the noise. What matters most from here is how I manage workloads: Morejon late in the season, then Ohtani, Snell and Glasnow.

PART 6

Simulating the 2027 season

This part plays my plan out game by game. Yamamoto goes every sixth day, with Snell, Ohtani, Glasnow, Sasaki, free agent Morejon and Wrobleski around him. Morejon builds up behind Sheehan, and the offense is exactly the 2026 Dodgers'. I played the season 2,000 times on the real 2027 schedule, with injuries, depth starters, call-ups and bullpen fatigue tracked in every game.

Table 8: How the simulation works.

Piece	How
Offense	2026 Dodgers exactly: every 2026 half-inning resampled, home and away (4.94 runs per game in 2026, 100-62)
Run prevention	Each half-inning's runs come from the 2026 league rate, scaled by the opponent's 2026 scoring and the pitcher's 2024-26 run value
Rotation	Projected per-start schedule, pitch counts and arm-IL risk from Part 5 (Morejon 45/55/62/70, then 75)
Depth starters	Stone, Zazueta, Ferris (from RosterResource's AAA arms, with my starter/reliever roles). First start up from the minors is 80 pitches, then 85
Bullpen (7)	Diaz and Scott split closing (the more rested one gets the save); Henriquez and Dreyer are the dawgs; Stewart and Klein in the middle; Sheehan the long man. Ohtani is a two-way player and doesn't count against the 13 pitchers
Reliever call-ups	Knack, Gervase, Casparius, Kopp, Hurt, Halvorsen, Miller, Serwinowski, Cousins (Knack, Casparius and Miller can go long)
Bullpen use	Closers in save or tie spots from the 9th on, dawgs in the 7th and 8th when the game is close, long men early or in blowouts. Outing length is drawn from 2023-25 league relief outings by reliever type, ending at 40 pitches; Sheehan always goes 2 innings
Reliever rest	Rested by pitches thrown (under 20: none, 20-29: 1 day, 30-39: 2, 40+: 3), never three days running
Injuries	Starters: Part 5 per-start arm risk plus the league non-arm rate. Relievers: per-outing model below. IL lengths drawn from real 2018-26 stints

Reliever injuries and workload

Relief injury risk comes from 93,651 relief outings (2021-26). Each outing's arm-IL risk depends on the reliever's recent work and history.

Table 9: Reliever arm-IL odds per outing. Base rate: 6.7 arm IL placements per 1,000 relief outings, plus 5.3 non-arm.

Factor	Odds ratio	95% CI	p
Pitched yesterday	0.89	0.70-1.13	0.34
Each outing in the last 7 days	0.91	0.82-1.02	0.099
+100 pitches in the last 7 days	2.38	1.44-3.92	0.00067
Each year of age	0.98	0.95-1.01	0.23
Each earlier arm IL (up to 3)	1.15	1.05-1.26	0.0033
Arm IL in the past year	1.66	1.33-2.06	<0.0001

Pitch volume is what matters: each extra 100 pitches in the past week multiplies the odds by about 2.4. Pitching back to back, or how many times he pitched, adds nothing once pitches are counted.

Results

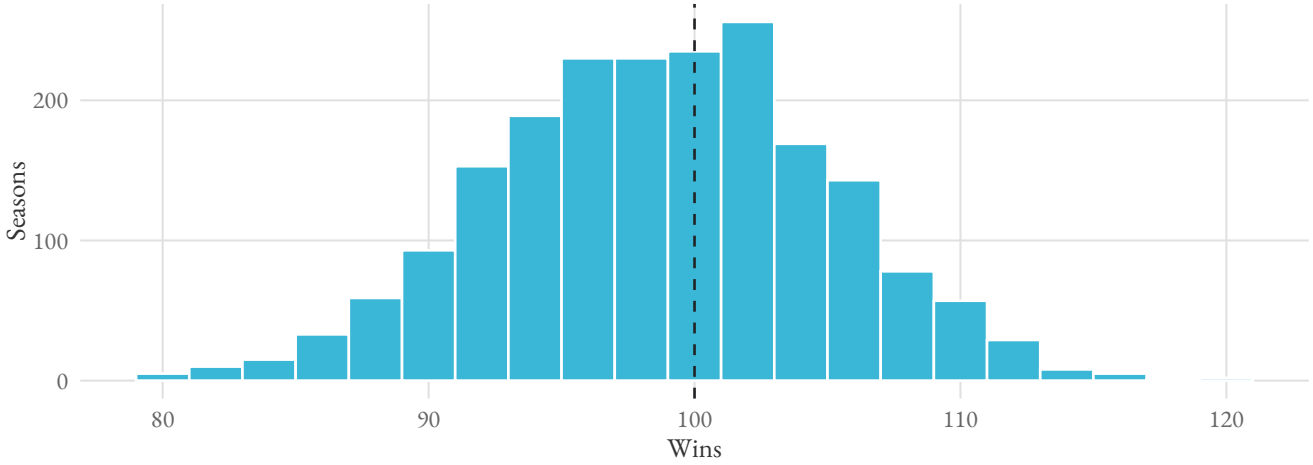


Figure 7: Wins across the simulated seasons. The dashed line is the 2026 Dodgers' 100 wins.

Table 10: Simulated 2027 season. Short-handed: no reliever on the active roster was rested, so a tired or emergency arm pitched.

Measure	Mean (10th-90th pct)
Wins	99.2 (91-107)
Runs scored	820.8 (771-872)
Runs allowed	680.3 (628-734)
Starts by depth starters	21.6 (5-39)
Rotation IL stints (arm / other)	2.2 / 2.1
Rotation days on IL	321.6 (67-598)
Bullpen pitches	10,244.1 (9,938-10,549)
Bullpen outings	497.6 (479-517)
Bullpen IL stints (arm / other)	3.9 / 2.6
Bullpen days on IL	497.0 (184-833)
Call-ups used	4.0 (2-6)
Short-handed relief outings	11.5

- **My plan wins about 99 games.** The median is 99. 77% of seasons reach 95 wins and 6% fall under 90. With the 2026 offense unchanged, the runs allowed (4.20 per game, against 3.70 in 2026) set the ceiling. Runs scored run slightly above 2026's 801 because a weaker staff means more bottom-of-the-ninth innings get played at home.
- **Injuries to starters are absorbed.** The rotation averages 4.3 IL stints and 22 depth starts. Stone takes most of them (11.0), then Zazueta (6.3) and Ferris (4.3). Each extra depth start costs a little (seasons with more depth starts win slightly less, correlation −0.20).
- **The bullpen is where injuries pile up.** It averages 6.5 IL stints a season (3.9 arm). In 11% of seasons it reaches 10 or more. About 4 call-ups get used, led by Knack (36 outings) and Gervase (32).

Table 11: Rotation and depth starters, per season (simulation averages). Planned starts: Yamamoto 30, the others 22 each.

Pitcher	GS	Pitches	Arm IL	Other IL	Days on IL	P(any IL)
Yamamoto	25.2	2,444	0.32	0.32	47	54%
Wrobleski	19.4	1,718	0.23	0.23	35	41%
Sasaki	19.3	1,714	0.22	0.26	36	42%
Glasnow	19.1	1,687	0.27	0.24	40	45%
Morejon	19.1	1,374	0.28	0.24	38	45%
Snell	19.1	1,682	0.31	0.25	41	48%
Ohtani	19.0	1,756	0.30	0.24	41	47%
Stone	11.0	935	0.14	0.14	22	27%
Zazueta	6.3	534	0.08	0.09	13	17%
Ferris	4.3	369	0.06	0.05	8	10%

Table 12: Bullpen usage and injuries per season (simulation averages). 3 in 4 days: outings that were his third in four days.

Reliever	Role	Outings	Innings	Pitches	Back-to-back	3 in 4 days	Arm IL	Other IL	P(any IL)
Henriquez	Opening Day	69	79	1,308	22	25	0.46	0.36	66%
Stewart	Opening Day	63	73	1,217	22	26	0.75	0.32	77%
Klein	Opening Day	60	69	1,161	19	20	0.62	0.33	74%
Dreyer	Opening Day	59	68	1,109	16	15	0.51	0.33	66%
Sheehan	Opening Day	47	88	1,452	3	1	0.37	0.26	53%
Diaz	Opening Day	46	48	810	7	4	0.34	0.22	49%
Scott	Opening Day	42	45	748	6	3	0.26	0.23	43%
Knack	Call-up	36	54	905	7	4	0.21	0.18	37%
Gervase	Call-up	32	36	616	9	9	0.20	0.16	34%
Casparius	Call-up	14	21	356	3	1	0.09	0.08	17%
Kopp	Call-up	10	12	199	3	2	0.06	0.05	10%
Hurt	Call-up	5	6	103	2	2	0.04	0.03	7%
Halvorsen	Call-up	2	2	36	1	1	0.02	0.01	2%

- **Henriquez is the busiest arm, and the middle men are close behind.** Henriquez makes 69 outings with 22 back-to-backs. Stewart (63) and Klein (60) absorb the middle innings and pitch nearly as often, and Dreyer makes 59. Splitting the closer job keeps Diaz (46 outings) and Scott (42) light, at about 7 back-to-backs each.
- **Sheehan's role holds up.** He averages 47 outings and 88 innings as the long man, 1,452 pitches, still well under his 1,960 as a 2026 starter. Most of his work comes in early exits and blowouts, not back-to-backs (3 a year).
- **Seven relievers is tight.** In about 11 outings a season nobody in the pen is rested, and a tired arm has to pitch. That is the cost of carrying seven starters. Morejon's short early starts add to it in April, but Sheehan's follow-ups cover those games.

> **Takeaway.** With the 2026 offense, my plan projects about 99 wins, close to 2026's 100. Rotation injuries are covered by Stone, then Zazueta, at a small cost. The bigger risk is the seven-man bullpen: about 6.5 IL stints a season, heavy back-to-back use for the setup men, and roughly a dozen short-handed outings. Pitch counts in the past week drive reliever arm injuries, so spreading pitches across the pen is the lever I have to watch. That is what I test with the shuttle in Part 7.

PART 7

Five plans, same season engine

I ran the same simulation, with the same random draws, for five plans. Each plan's bullpen is whatever is left of the 13-pitcher staff after its rotation. Ohtani is a two-way player and doesn't count against the 13. Whenever free agent Morejon starts, he builds up with Sheehan behind him.

Table 13: The five plans. Core six relievers: Diaz, Scott, Henriquez, Stewart, Dreyer, Klein.

Plan	Rotation	Bullpen
Ace + 6 (Morejon)	Yamamoto every 6th day; Snell, Ohtani, Glasnow, Sasaki, Morejon (FA), Wrobleski	7: core six + Sheehan
Ace + 5 (Morejon)	Yamamoto every 6th day; Snell, Ohtani, Glasnow, Sasaki, Morejon (FA)	8: core six + Sheehan, Wrobleski
Strict 6-man	Yamamoto, Snell, Ohtani, Glasnow, Sasaki, Morejon (FA)	8: core six + Sheehan, Wrobleski
Strict 5-man	Yamamoto, Snell, Ohtani, Glasnow, Sasaki	8: core six + Sheehan, Wrobleski (Morejon not signed)
Ace + 6 (Sheehan)	Yamamoto every 6th day; Snell, Ohtani, Glasnow, Sasaki, Sheehan, Wrobleski (Morejon not signed)	7: core six + Knack

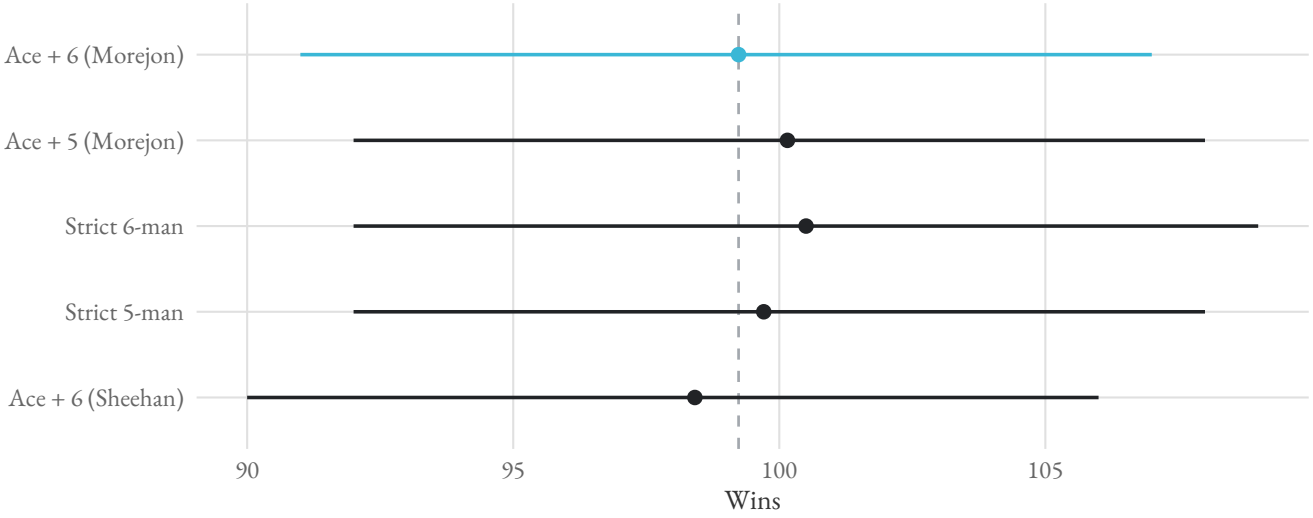


Figure 8: Wins by plan: mean (point) and 10th-90th percentile range across 2,000 seasons each. Dashed line: my plan's mean.

Table 14: Season averages by plan (2,000 seasons each). IL counts include arm and other stints. Top-3 RP and B2B: outings and back-to-backs for the three busiest relievers combined. Short-handed: relief outings with no rested arm available.

Plan	Wins	95+ wins	Runs allowed	Depth starts	Rotation IL	Bullpen IL	Top-3 RP outings	Top-3 B2B	Short-handed
Ace + 6 (Morejon)	99.2	77%	680	22	4.3	6.5	222	66	11.5
Ace + 5 (Morejon)	100.2	82%	675	26	4.3	6.5	210	58	2.6
Strict 6-man	100.5	83%	676	26	4.4	6.6	211	58	2.7
Strict 5-man	99.7	79%	681	33	4.7	6.5	207	56	1.5
Ace + 6 (Sheehan)	98.4	74%	696	22	4.2	6.5	224	68	9.2

- **The gaps are small: about 2.1 wins from top to bottom.** With 2,000 seasons each, the simulation noise on a plan's average is about 0.15 wins, so differences of half a win or more are real within the model. They are still small next to the model's own assumptions.

- **The strict 6-man wins the most (100.5), then ace + 5 (100.2).** Both put more of the season on the best starters and leave Wrobleski in an eight-man bullpen.
- **Without Morejon in the pen, the strict 5-man falls back to 99.7.** Its eight-man bullpen is the best rested (1.5 short-handed outings), but five starters on regular rest break down more often: 33 depth starts, the most of any plan, and 4.7 rotation IL stints.
- **My plan, ace + 6 with Morejon, comes in at 99.2.** It loses about 1.3 wins to the 6-man, 0.9 to ace + 5 and 0.5 to the 5-man. The cost isn't in the rotation. It's the seven-man bullpen: 11 short-handed outings a season, against 3 or fewer in the three plans with eight relievers, plus more back-to-backs for the top relievers (66 against 58 in the 6-man).
- **What it buys is a healthier, steadier rotation.** It has the fewest depth starts of any plan (22, against 33 in the 5-man) and fewer rotation IL stints (4.3 against 4.7). In wins, that doesn't make up for the thinner bullpen.
- **Bullpen injuries barely move between plans (about 6.5 a season in every plan).** Fewer relievers each pitch more, but more relievers means more arms that can get hurt. The two roughly cancel.
- **Signing Morejon is worth about 0.8 wins inside the ace + 6 frame.** Ace + 6 with Sheehan in the rotation instead of Morejon is the weakest plan (98.4 wins, 696 runs allowed). Sheehan is closer to average as a starter, and Knack becomes the seventh reliever.

My bullpen roles, optionable depth and a shuttle

All plans above use my bullpen roles. For ace + 6 with Morejon, I tested two more versions of the depth behind it. **Optionable depth** drops Knack, who is out of options, from the call-ups. **Shuttle** keeps that optionable-only depth and adds a rule: before any game where fewer than four relievers are rested, up to two tired optionable arms (Klein or a call-up, never the closers, the dawgs or Sheehan) go down for the required 15 days, and fresh arms come up.

Table 15: Ace + 6 with Morejon: depth versions (2,000 seasons each).

Version	Wins	Runs allowed	Short-handed	Bullpen IL	Top-3 B2B	Call-ups used	Shuttle moves
Full depth (with Knack)	99.2	680	11.5	6.5	66	4.0	0.0
Optionable depth only	99.4	683	11.6	6.6	67	4.2	0.0
Optionable depth + shuttle	99.6	682	6.1	6.7	62	6.6	12.5

- **Losing Knack changes wins by +0.2.** Short-handed outings go from 11.5 to 11.6. Gervase and Casparius take over as the first call-ups.
- **The shuttle makes 12 moves a season and cuts short-handed outings to 6.1.** Wins change by +0.2 against optionable depth without it, and bullpen IL stints by +0.0.
- **The 15-day option rule is the limit.** Each arm sent down can't come back for two weeks, so the shuttle reaches deeper into the list: 6.6 call-ups used a season, against 4.2 without it.

Yamamoto on his 5th day before an off day

My rule sends Yamamoto on the seventh day whenever his sixth is an off day. This version flips that: if the sixth day is an off day, he goes on the fifth, on four days of rest, and the off day goes to the other six. Everything else is ace + 6 with Morejon building up behind Sheehan.

Table 16: Ace + 6 with Morejon: Yamamoto on the 7th day after an off day (my rule) or the 5th day before it. Top rows are the planned schedule; the last four are season simulations (2,000 each).

Measure	My rule (7th day)	Early (5th day)
Yamamoto starts	30	32
Yamamoto starts on 4 days of rest	0	7
Yamamoto pitches	2,910	3,104
Yamamoto starts past his 2026 pitch total	1	3
Yamamoto expected arm-IL stints	0.37	0.42
Morejon / Wrobleski starts	22 / 22	21 / 21
Staff expected arm-IL stints	2.21	2.23
Starter runs, net of injuries	+51.9	+52.5
Simulated wins	99.23	99.67
Simulated runs allowed	680	678
Short-handed relief outings	11.5	10.6
Yamamoto P(any IL), simulated	54%	56%

- **Two more starts for my best pitcher.** Yamamoto goes from 30 to 32 starts. 7 come on four days of rest, April 11 through August 18. Morejon and Wrobleski each give up a start, and the other five get a little more rest.
- **It wins about 0.4 more games** (99.7 against 99.2), with 2 fewer runs allowed. Yamamoto's deeper starts also take a little work off the bullpen: short-handed outings drop to 10.6.
- **The injury cost is small.** Four days instead of five is the rest gap Part 2 found barely matters. The bigger change is volume: 194 more pitches mean he passes his 2026 total 2 starts earlier. His expected arm-IL stints go from 0.37 to 0.42, and the whole staff's from 2.21 to 2.23.
- **It closes about half the gap to ace + 5** without changing the roster, and it's worth more than the shuttle. The two could be combined, since one changes the rotation and the other the bullpen.

> **Takeaway.** In the season simulation, my ace + 6 with Morejon is a sound plan, but not the one that wins the most. Most of its cost comes from carrying seven starters and only seven relievers. If I keep the every-6th-day rule for Yamamoto, ace + 5 with Morejon recovers most of the gap (about 0.9 wins) by putting Wrobleski in the bullpen. Inside ace + 6, I have two cheap levers: start Yamamoto on his 5th day before an off day (+0.4 wins) and shuttle fresh arms up from AAA when the pen is short (+0.2).

APPENDIX

Data, methods and checks

Data

- **Starts.** Retrosheet play-by-play 2010-2025 (first pitcher for each team in each game, pitches summed from pitch counts) and Statcast 2026. Rotation pitchers: 8+ starts that season with a median start of 60+ pitches, which drops openers. Days of rest = days since the pitcher's previous start, minus one. Starts with a relief outing in between are left out.
- **Turn.** Team games since the pitcher's previous start. On a 5-game turn the club didn't skip or push him, so 4, 5 or 6 days of rest came from off days alone.
- **IL placements.** MLB StatsAPI transactions 2009-2026, pulled month by month (one season-long query misses placements). 6,525 pitcher placements 2010-26, 3,158 of them to the arm. Retroactive dates are used when given. 27 placements missing from the feed (Yamamoto's June 2024 triceps stint among them) were rebuilt from the 15-to-60-day transfer, dated the day after the pitcher's last appearance. Off-season-dated placements are dropped because several are misdated. COVID and "recovering from surgery" stints are excluded.
- **Arm.** Elbow (UCL, flexor, pronator, ulnar nerve), shoulder (rotator cuff, labrum, capsule, AC, pectoral, trapezius) and other arm (forearm, biceps, triceps, lat, teres).

Table 17: Rotation starts by season. Arm IL = arm placements effective before the next appearance (21 days at most).

Season	Starts	Pitchers	Arm IL	Per 100	P/GS	4 days	5 days	6+ days
2010	4,620	195	30	0.65	97.6	51%	33%	11%
2011	4,594	195	40	0.87	97.7	49%	34%	12%
2012	4,548	198	43	0.95	95.7	48%	35%	13%
2013	4,511	197	45	1.00	96.2	50%	31%	14%
2014	4,568	197	34	0.74	96.4	48%	34%	13%
2015	4,516	206	41	0.91	94.0	48%	34%	13%
2016	4,498	203	41	0.91	93.5	48%	32%	15%
2017	4,523	210	47	1.04	92.5	47%	33%	16%
2018	4,358	193	54	1.24	90.5	39%	39%	17%
2019	4,293	196	43	1.00	90.0	41%	36%	18%
2020	1,216	116	9	0.74	86.7	43%	31%	16%
2021	4,355	208	59	1.35	85.4	34%	38%	22%
2022	4,433	207	67	1.51	87.1	38%	37%	20%
2023	4,306	206	50	1.16	88.5	33%	43%	19%
2024	4,298	197	60	1.40	88.1	33%	44%	18%
2025	4,399	207	62	1.41	87.3	30%	48%	18%
2026	4,355	203	64	1.47	87.0	30%	46%	19%

Methods

- **Injury models.** Logit at the start level. Outcome: arm IL placement effective on or after the start date and before the next appearance (21 days at most). Controls: previous start's pitch count (held at 90 in the 2027 projection, because pitchers who are already hurt get pulled early, which makes a low count look risky), age and age squared, career arm-IL count (capped at 3), arm IL in the last 365 days, month and season. Standard errors clustered by pitcher. The start's own pitch count is left out because an injury cuts the start short.
- **Designs.** All starts. Schedule-set: on-turn starts only (5-game turn, rest 4-6; 6-game turn, rest 5-7). Same pitcher-season: conditional logit within each pitcher-season that had an event. Workload jump: cumulative MLB pitches this season above last season's total (pitchers with 800+ pitches the year before, 2021 dropped because 2020 was short).
- **Performance.** Outcome minus the pitcher-season mean, regressed on rest bucket and month. K-BB% and wOBA

are weighted by batters faced. Starts followed by an arm IL are left out.

- **2027 projection.** A 2018-26 logit with the same terms plus the workload-jump flag gives each start's risk at four days of rest. The rest odds ratio is applied on top: schedule-set 2017-26 (main), fitted, or none. Run value: 2024-26 run value per 100 pitches as a starter, shrunk with 3,000 pitches of average. Morejon (a free-agent signing with no full-length starts) keeps his shrunk 2024-26 relief value in the main case. Alternatives are a 0.6 run cut, the average converter (+0.35) and his raw relief value. A replacement-level starter is −0.6 runs per 100 pitches. Pitches per start: 2026 averages (Yamamoto 97, Ohtani 92, Morejon 75, others 88).

Table 18: Rest odds ratios (vs. 4 days) used in the 2027 projection.

Days of rest	Schedule-set 2017-26 (main)	Fitted 2018-26	None
4	1.00	1.00	1.00
5	0.89	0.80	1.00
6	0.79	0.75	1.00
7-9	0.79	0.63	1.00
10+	1.00	1.01	1.00

Limits

- **Few events.** Fewer than 800 arm placements follow a start. The six-plus-day estimates rest on a few dozen events, so intervals are wide.
- **The IL feed isn't perfect.** Some placements are missing or misdated, and about 2% of activations have no matching placement. Arm injuries that never reach the IL, or that show up between starts after a bullpen session, are missed.
- **Seven-day rest is rare.** A strict 7-day turn barely exists in MLB, so the effect of the rest my plan gives the other six is projected from 6-day starts.
- **Run value inputs are rough.** Sasaki, Wrobleski and especially Morejon as a starter carry wide uncertainty, and Ohtani's two-way workload isn't modeled.
- **The simulation is simple by design.** The offense is fixed at 2026 with no hitter injuries. Relief outing lengths follow league patterns by reliever type, but a manager's in-game choices (matchups, warm-ups that never pitch) aren't modeled. Starters leave on pitch count, not on how they're pitching. Defense is folded into each pitcher's run value, and opponents' 2027 rosters are their 2026 scoring.
- **Roster moves are assumptions.** MLB's data still lists Morejon with the Padres. I treat him as a Dodgers free-agent signing at his 2024-26 level. Morejon's bullpen value is not credited in the plans where he relieves.